Venkata Lakshmi Parimala Pasupuleti

Software Engineer & AI Builder · Python · PHP · LangGraph · RAG · SQL · AWS · Agentic AI · PropTech · Production AI · M.S. Data Analytics Engineering

+1-703-261-5410 | venkatalakshmiparimala@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | github.com/venkatalakshmiparimala/AgentProbe | Portfolio | Virginia, USA — Remote

PROFESSIONAL SUMMARY

Software engineer and AI builder with almost 10 years shipping production systems. Built **AgentProbe**: 3-agent LangGraph runtime with multi-provider LLM routing, 5-metric eval engine, CI/CD regression gates, and React observability dashboard. Built **Leasel**: agentic RAG pipeline (93.6% precision, 1,240 PDFs, deterministic rules engine). Built agentic drone mission planner at Magizhs (LangGraph, RAG over FAA Part 107). At Entrata: financial data pipelines (HAP, HUD, tax credits, 1.6M records), 200+ SQL optimizations, Docker microservices (50% scalability), 100% PHI compliance. Currently GTA at GMU - mentoring 600+ students in Python and database design. M.S. Data Analytics Engineering, GMU (GPA 3.87).

WORK EXPERIENCE

George Mason University — College of Engineering and Computing

Aug 2025 – Present

Graduate Teaching Assistant — Python · Database Design · SQL · MS Access · Excel

- Mentoring **600+ students** across multiple sections of MIS 303 in Python programming (data structures, OOP, automation scripting) and database design (ER modeling, normalization, SQL query writing, MS Access, Excel data analysis); developing supplementary exercises, grading with rubric-aligned feedback, and coordinating content delivery with lead instructor.

Magizhs Tech Solution

May 2025 – Aug 2025

AI/ML Engineer Intern — Agentic AI · LangGraph · RAG · FAA Compliance · Drone Mission Planning

- Built an **agentic drone mission-planning system** orchestrated as a LangGraph stateful graph (parse → plan → retrieve → validate → conditional routing); RAG layer over FAA Part 107 corpus grounds every compliance check in cited source regulations; deterministic validation guardrails enforce altitude limits, battery budgets, and no-fly-zone checks — blocking non-compliant missions before MAVLink/QGroundControl export.

George Mason University

Aug 2024 – Aug 2025

Research Associate — ETL Pipelines · ML Modeling · Anomaly Detection · Analytics

- Built multi-source ETL pipelines (Python, R, 190+ countries) achieving **$R^2 = 0.977$**; applied Isolation Forest anomaly detection across 3 research pipelines; designed analytics dashboards (Tableau, ggplot2, Quarto) driving **40% improvement** in decision comprehension for faculty leadership.

Entrata — PropTech / Real Estate SaaS

Sep 2018 – Aug 2024

Senior Software Engineer — Full-Stack · Financial Data Pipelines · Production AI · Compliance · AWS · Microservices

- Shipped 20+ full-stack production modules (PHP, Python, PostgreSQL, JavaScript, RabbitMQ) across 120+ Agile sprints — **100K+ MAU, 1M+ records, 99.5% uptime**; managed full financial operations stack: **HAP allocation, HUD payment processing, rent schedules, gross rent, tax credit calculations, ledger management, rental subsidies, and income eligibility** across 100K+ unit portfolios — zero compliance violations.
- Shipped Entrata's **first production AI pipeline on 1.6M financial records** (Python, PyTorch, OpenAI API); **optimized 200+ SQL queries** (30% gain) via execution plan tuning, indexing, partitioning, caching, triggers, audit tables; led Docker microservices redesign (**50% scalability, 30% deployment reduction**); achieved **100% PHI compliance** with Libsodium-encrypted AWS S3 migration.
- Created **56 state-specific housing certification documents** with AWS S3 integration; eliminated 80% manual workload via Python/Bash automation; built debugging tools reducing resolution time 40%; **300+ code reviews, 5+ engineers mentored**.

Entrata India

Jan 2017 – Sep 2018

Associate / Trainee Software Engineer — Full-Stack · PHP · PostgreSQL · Data Pipelines · AWS S3

- Built full-stack property management modules (PHP, PostgreSQL, JavaScript); designed SQL import scripts for 1.6M HUD/Tax Credit records; migrated client documents to AWS S3 resolving 90% of legacy space constraints; participated in Agile sprints, code reviews, and compliance module development.

PROJECTS

AgentProbe: Multi-Agent AI Toolkit — Eval Engine, Observability & CI/CD Gates

AGENTIC AI · LLM EVALUATION

Python · LangGraph · LangChain · Claude API · OpenAI · Groq · Ollama · FastAPI · ChromaDB · Docker · GitHub Actions · React — github.com/venkatalakshmiparimala/AgentProbe

- 3-agent LangGraph runtime (Orchestrator, RAG Tracer, Auto Fix) with tool calling, inter-agent state handoffs, and multi-provider LLM routing (Claude, OpenAI, Groq, Ollama); **5-metric LLM-as-judge eval engine** with CI/CD regression gates on every commit; React observability dashboard tracking per-run metrics, latency, and retrieval drift automatically.

Leasel: Agentic RAG for Affordable Housing Compliance (1,240 Legal PDFs)

PROPTECH · AGENTIC RAG

Python · LangGraph · ChromaDB · Deterministic Rules Engine · LLM-as-Judge · Pass@k Eval — github.com/venkatalakshmiparimala/Lease_complaince_rag

- **93.6% precision, 87.9% F1** across 10 compliance rules on 1,240 residential lease PDFs; deterministic rules engine keeps all compliance outputs fully auditable; pass@k eval harness (pass@1 81.7%, pass@3 88.3%) with CI/CD regression gate.

AI on FHIR: NLP-Driven Medical Records Query System

NLP · HIPAA

Flask · LLMs · FHIR · React · Docker · spaCy · BERT · Embeddings — github.com/venkatalakshmiparimala/AI_on_FHIR

- Reduced manual clinical retrieval time ~60% by building a Flask + React NLP platform translating natural language queries (EN/ES) to FHIR APIs using LLMs, BERT, and embeddings; HIPAA-compliant RBAC enforced across all structured medical records.

EDUCATION

M.S. Data Analytics Engineering — George Mason University — May 2026 · GPA 3.87

ML systems, AI pipelines, data engineering, statistical modeling, anomaly detection, data governance

B.Tech Computer Science & Engineering — JNTU Anantapur — 2013 – 2017

TECHNICAL SKILLS

Languages: Python, PHP, JavaScript, TypeScript, SQL (expert — 200+ production optimizations, schema design, execution plan tuning, indexing, partitioning, triggers, audit tables), PL/SQL, R, jQuery, AJAX, HTML/CSS, Bash

AI & Agents: LangGraph, LangChain, Agentic AI, Multi-Agent Orchestration, Tool Calling, RAG Pipelines, Vector Databases, ChromaDB, Semantic Search, Embeddings, LLM Evaluation, LLM-as-Judge, Eval Frameworks, Prompt Engineering, Agent Observability, Claude API, OpenAI API, Groq, Ollama, NLP, BERT, Hugging Face, FastAPI, REST APIs

ML & Data: PyTorch, scikit-learn, XGBoost, Random Forest, Isolation Forest, Feature Engineering, Anomaly Detection, Statistical Modeling, Pandas, NumPy, ETL/ELT Pipelines, Data Normalization, Schema Design, Tableau, Power BI, ggplot2

Cloud & Infrastructure: AWS (S3, EC2, RDS, Glue), Docker, Microservices, CI/CD (GitHub Actions), PostgreSQL, MySQL, Oracle, RabbitMQ, Libsodium, Salesforce, Clay, n8n, Git, Linux, Agile/Scrum, Planning Poker